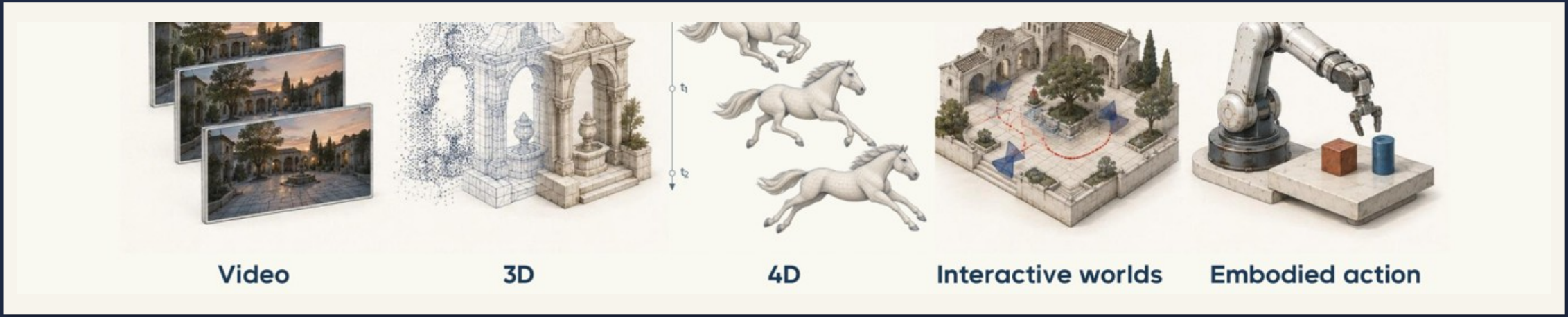
Paper Review • arXiv:2611.10771 • 2026-11-07

WorldFoundry

Towards Unified Infrastructure for World Intelligence

Closure defines world models; unified infra makes them reproducible and comparable

OpenEnvision • Juanxi Tian, Jiadong Hou, …, Ming-Hsuan Yang, Felix Heide, Wei Xiong, Enze Xie, Wanhua Li



TL;DR: define world models by closure; measure with unified infra

World model ≠ photorealism. The criterion is **closure**: interventions leave traces in a persistent state, and observations correct that state. WorldFoundry uses manifests, artifact contracts and evidence scorecards to place heterogeneous world models in one reproducible, traceable inference–evaluation pipeline.

270	74	62	5
Model entries	Benchmarks	Reg. metrics	Modalities

Five modalities: video • 3D • 4D • interactive worlds • embodied action. Every model runner is verified end-to-end on a documented reference workflow (run-ready ≠ performance claim).

Problem: bottlenecks lie in infra & evaluation, not just algorithms

① Fragmented workflows

- Video generators, neural scene representations and embodied policies each cover one piece
- Incompatible representations / runtimes / interfaces / eval protocols
- → Hard to compose capabilities, reproduce results, or measure progress

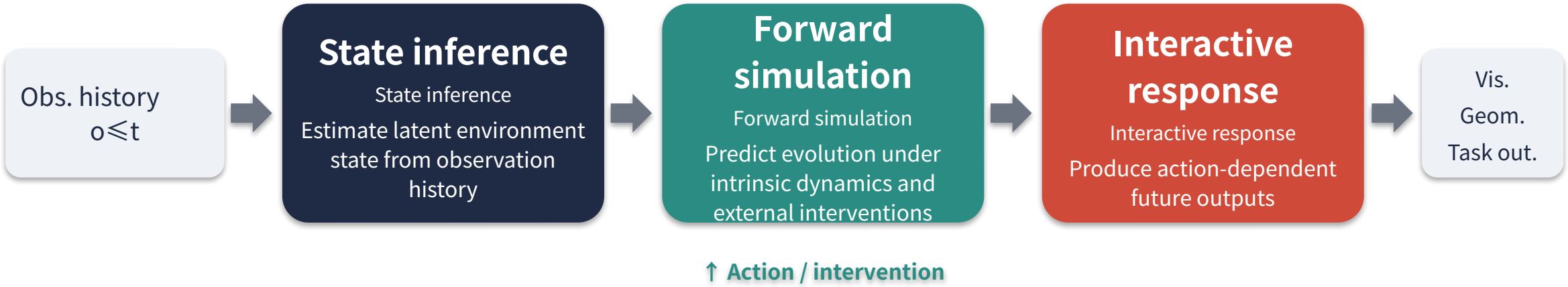
② Perception-centric evaluation

- Visual plausibility reflects only the "observation decoder"
- Neither necessary nor sufficient for persistent state
- Physics/world benchmarks repeatedly expose gaps (VBench-2, VideoPhy-2, WorldModelBench, WorldArena)

Result: the line between "foundation generative models" and "generative world models" blurs; dynamics, controllability and interactive utility cannot be evaluated reproducibly

Definition • 1

Minimal functional abstraction: infer state → simulate → respond



Functional, not architectural: a shared interface; systems need not implement all three operators — model families differ in focus

Static 3D

Persistent state inference + geometry decoding

Video models

Implicitly learned state transitions + observation generation

4D methods

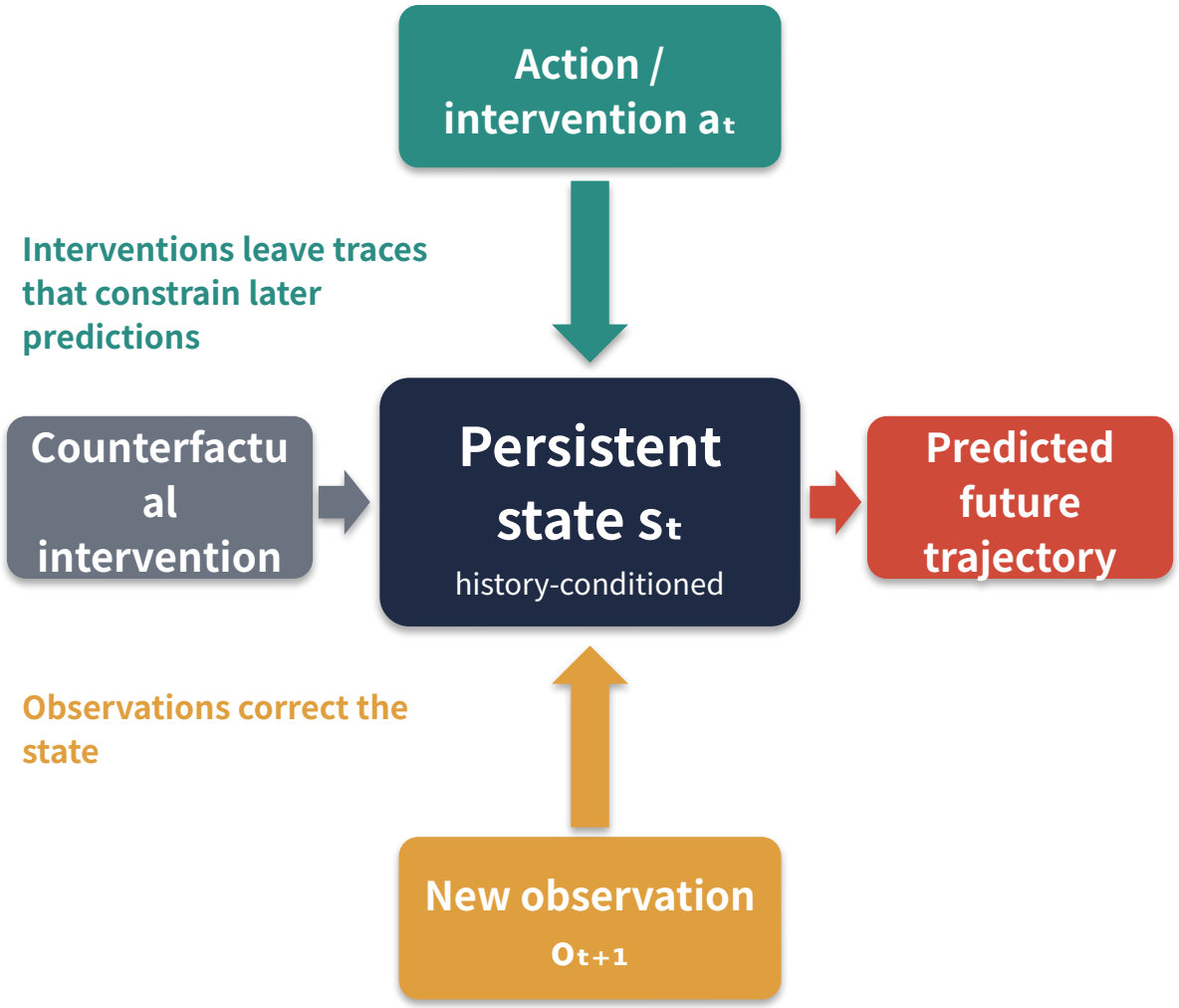
Explicitly represent time-varying structure

Embodied models

State estimation + action prediction → decisions

Definition • 2

The criterion is closure, not visual plausibility



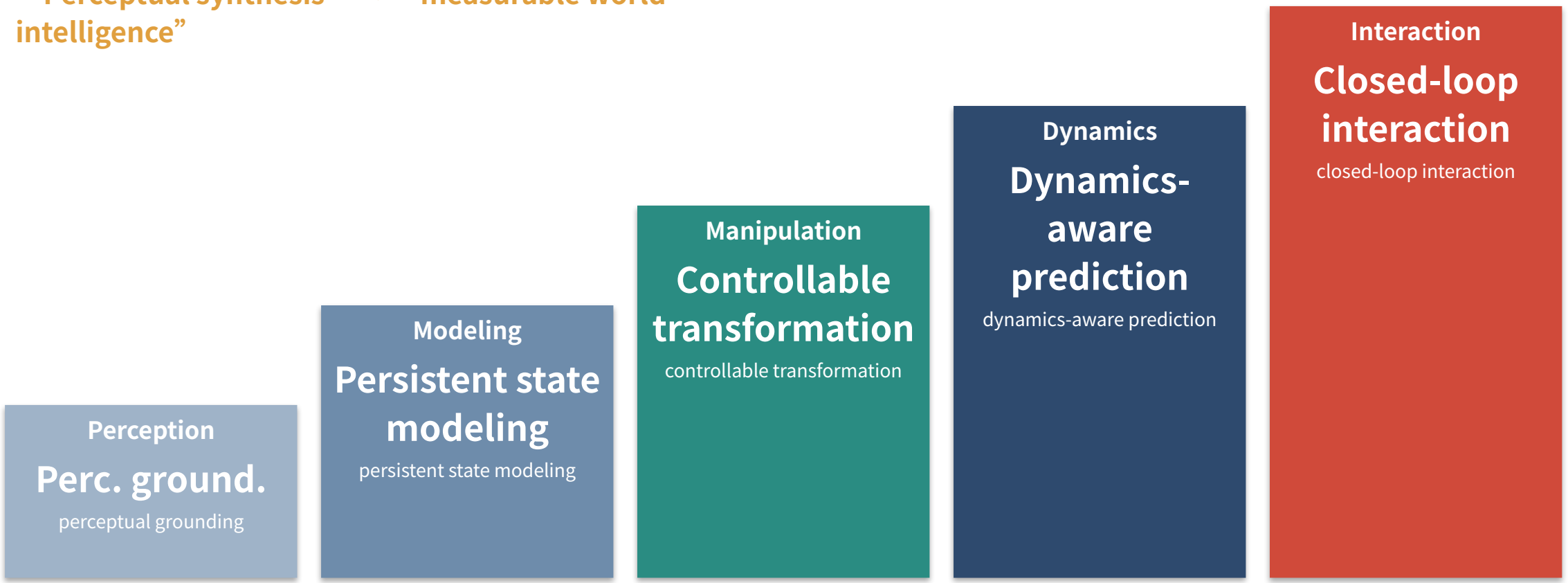
Foundation generative model	Generative world model
Synthesizes / transforms observations	**Generation coupled to an evolving state**
No persistent state needed	**Infers and maintains persistent state**
Interventions need leave no trace	**Interventions leave traces, constrain future**
Observations not written back to state	**Absorbs observations, corrects state**
Visual plausibility suffices	**Criterion: closure**

"It is closure, not plausibility, that draws a principled boundary."

Definition → capabilities: from perceptual synthesis to closed loop

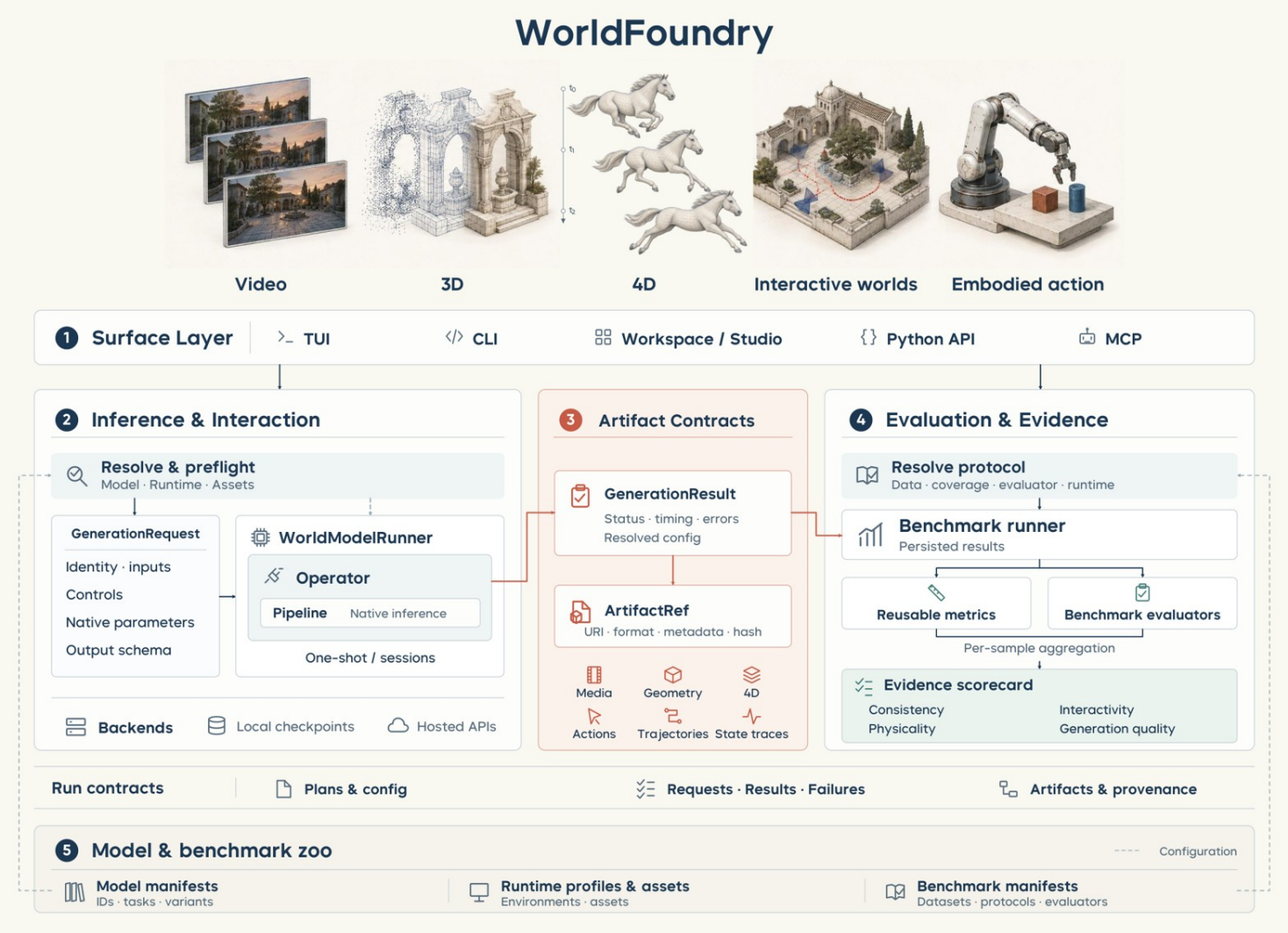
“Perceptual synthesis” → “measurable world intelligence”

Use: build “capability profiles” of heterogeneous systems — compare by implemented operators and supported capabilities, not architecture.



System

WorldFoundry: five layers linking discovery → run → artifact → eval



1 Surface Layer

TUI • CLI • Studio • Python API • MCP

2 Inference & Interaction

preflight → Request → Runner → native pipeline; one-shot / session

3 Artifact Contracts

GenerationResult + ArtifactRef(URI • hash)

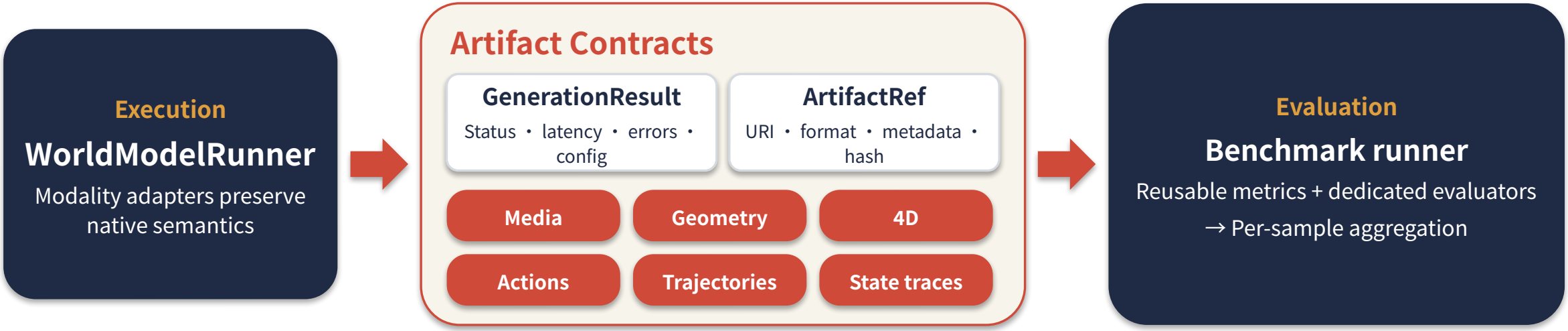
4 Evaluation & Evidence

Protocol → benchmark runner → metrics → scorecard

5 Model & Benchmark Zoo

Model / runtime / benchmark manifests

Artifact contract: decoupled execution/eval, traceable claims



Evidence scorecard: four dimensions

Consistency	Physicality	Interactivity	Generation quality
Consistency	Physicality	Interactivity	Generation quality

Benefit: execute once, evaluate many times; claims are auditable and re-checkable (each artifact carries a hash and resolved config)

Evidence tiers: “runs” and “runs well” are reported separately

Catalog registration: model / benchmark / metric entries and their manifests

Declared support

Declared support

Passes reference workflow end-to-end; execution evidence recorded

Run-ready

Operational readiness

Eval scores on persisted artifacts, traceable to those artifacts

Validated performance

Validated performance

Current catalog (abstract figures)

270 Model entries

74 benchmarks

62 Reg. metrics

All model runners verified via end-to-end reference workflows = run-ready, not a performance claim

Verdict: solid definition & engineering; quantifying closure is open

Strengths

- Operational definition: intervention traces + observation correction point to evaluating state persistence and counterfactual response
- Sound engineering abstractions: manifest → request → artifact → scorecard; reproducible and traceable
- Broad coverage: unified dispatch across five modalities; MCP support eases agent use
- Evidence tiers: separates “integrated” from “evaluated”

Limitations / Questions

- No substantive cross-model evaluation results in the available material
- Not yet shown how the closure criterion becomes concrete metrics
- Can video models and embodied policies fairly share interactivity/physicality metrics?
- Implicit-state persistence testable only via behavioral probes; long-term value depends on maintenance and community

Takeaways

For embodied AI: compare “world model + policy” under one contract

01 Unified framework

Embodied action models share execution and evaluation contracts with video / 4D / interactive worlds; Actions, Trajectories and State traces are first-class artifacts

02 Selection criteria

“Intervention traces + observation correction” match what embodied AI needs from world models (predicting action outcomes, supporting planning)

03 Research practice

Submit results with manifests and traceable artifacts, turning isolated demos into cumulative, auditable evidence

Take-home: moving from “perceptual synthesis” to “measurable world intelligence” requires closed-loop evidence and reproducible infrastructure